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DATE	25-Feb-2011	
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Revision	Date	Reason for Issue	Originator
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Revisions to this document are highlighted and are summarised below:

Section 5.1.1, Steam "PSV Setting" pressure changed to "Design" for clarity

Section 5.1.1, Silica in Steam corrected from 0.01ppm max to 0.02ppm max based on silica content in Clean Condensate and Demineralised Water.

Section 5.6.1, Fuel Gas supply pressure at consumer B/L revised as per ATI-0080.

Section 5.6.2, Sulphur content in Fuel Oil retained as in BEDD RevA4 at 2500pmm (ATI-0063)

Section 5.6.3, HSFO Supply header normal pressure at CPP B/L reduced from 4.0 kg/cm2g to 3.0 kg/cm2g based on Hydraulics.(refer IAN 423 rev4)

Section 5.6.4, Gas Turbine Fuel Supply Pressure (Min/Norm/Max) at CPP Battery Limit revised based on over all hydraulics (PDRP4280-8110-CA-000-0005 revF1)

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1 Project Description

1.1 Client: Indian Oil Corporation Limited – (IOCL)

1.2 Plant Name: Paradip Refinery Project (PDRP)

1.3 Plant Location:

Country	:	India
State	:	Orissa
District	:	Jagatsinghpur
Town/City	:	Abhaychandrapur, Paradip
Geographic bearing of site	:	20° 15' 20" latitude
	:	86° 36' 00" longitude
Site altitude	:	Minimum 3.91 m above Indian mean sea level (IMSL) which corresponds to the site recorded flood level

1.4 Project Type

This is a grass root Refinery project.

1.5 Units:

The Refinery is comprised of the following Units;

Atmospheric Vacuum Distillation, Naphtha Hydrotreaters, CCR Reformer, LPG Treating, ATF Treater, Diesel Hydrotreater, VGO Hydrotreater, Fluid Catalytic Cracker, Propylene Recovery, Delayed Coker, Alkylation, Butane Isomerisation, Hydrogen Generation, Amine, Sour Water Stripper, Sulphur Recovery, Spent Acid Regeneration, Flue Gas Desulphurisation, FCC Light Naphtha Treatment, FCC LPG Treatment, Straight Run LPG Treatment, Utilities, Offsites

1.6 The system of measurements shall be Metric as follows:

UNITS OF MEASUREMENT		
Parameter	*Preferred Unit*	*Alternative*
Area	m2	mm2
Composition - Gas or Vapor	vol% or mol%	ppmv
Composition - Liquid	wt%	ppmw
Enthalpy, Entropy	kcal/kg	
Flow (gas)	Nm^3/hr	kg/hr
Flow (liquid)	m^3/hr	kg/hr (for < 1TPH), TPH (for>1TPH)
Flow (steam)	kg/hr	
Force	N	
Fouling resistance	m^2 hr °C / kcal	

Furnace draft	mm of WC	
Heat	kcal	MMkcal/hr
Heat rate	10^6kcal/hr or MM kcal/hr	
Heat Transfer coefficient	kcal/hr-m^2-°C	
Length, Level	mm	m
Liq density	kg/m^3	
Liq relative density	sp gr T°C/15.6°C	
Mass	kg	
Pipe size / diameter	mm (DN)	
Plant Capacity	TPH	m^3/hr, KTPA, MMTPA
Power	kW	
Pressure - absolute	kg/cm² abs	
Pressure - gauge	kg/cm² g	
Sound Power	dB(A)	
Sound Pressure	dB(A)	
Specific Heat	kcal/kg°C	
Standard liquid	Sm^3/hr at 15.6°C	
Standard vapour	Nm^3/hr at 0°C & 1.033 kg/cm^2a	
Storage tank pressure	mm of WC	
Structural Stress	N/m2	
Temperature	°C	
Thermal conductivity	kcal/hr-m-°C	
Time	hr	sec, min, D
Vacuum	mm of Hg, mm WC	
Vap flowing density	kg/m^3	
Velocity	m/s	km/hr
Viscosity - Dynamic	cP	
Viscosity - Kinematic	cSt	
Volume	m3	

2 Applicable Codes, Standards, and Regulations

The Project Specifications and Standards are primarily prepared on the basis of Indian regulations, codes and standards. Where Indian codes are not applicable, international American /European/ British /other codes will be utilised accordingly after IOCL approval. In any areas of conflict, the order of precedence will be as follows (commencing with the highest):

1. National and Local Rules and Regulations
2. National Codes and Standards
3. Licensor Specifications and Standards
4. Engineering Standards and Specifications developed for the Project
5. International Codes and Standards

IOCL will not impose their corporate standards

In general, Euro Norms will apply for Electrical and Instrumentation, API, ASME and ANSI Codes will apply to Piping and Equipment.

2.1 International/Indian National/Project Specific Engineering Standards,

Project specific Engineering Standards/Specifications, based on FW Standards, shall be used where these are required to supplement International Standards and Indian National Standards.

The following listings of applicable Standards/Specifications will be prepared for the Project;

Engineering Standards and Specifications Index (PDRP-8820-IN-0001)

International Codes and Standards Index (PDRP -8820-IN -0002)

Indian National Codes and Standards Index (PDRP -8820-IN-0003)

The Indian Oil Industry Safety Directorate (OISD) codes and standards, in particular related to equipment spacing, are mandatory and take precedence in the event of conflict with other project standards

Bureau of Indian Standards (IS) standards are the governing Standards for both Civil and Electrical disciplines.

2.2 Government Acts, Regulations applying on general basis to the Paradip Project

The Paradip Refinery Project shall comply with the following requirements of Indian authorities:

(a) Factories Act; (As specific to Paradip)
(b) Indian Petroleum Rules;
(c) Tariff Advisory Committee Guidelines;
(d) Liquid effluent discharge, as per Minimal National Standards for liquid effluents and air emissions conforming to Pollution Control Board Standards;
(e) Civil Aviation Rules;
(f) Indian Boiler Regulations;
(g) Indian Electricity Rules;
(h) Requirement of Chief Controller of Explosives;
(i) Requirement of Town & Country Planning Department;
(j) Requirements of other authorities concerned with the Project
(k) All statutory provisions of India / Govt of Orissa

2.3 As a guide, design and construction shall conform to the latest edition of the following international, national and project basic standards, codes and/or regulations:

Pressure Vessels: ASMEVIII DIVI/II

Boilers: INDIAN BOILER REGULATIONS (IBR)

Buildings: IS/NBC

Structural: IS

Electrical: IS, IEC, Indian Electricity Rules

Sanitary: IS

Aircraft Warning: Airport Authority of India (Bhuvaneshwar, MOD)

Safety: OISD, Respective codes for individual equipment, Standard Engineering Practices, Project Standards., Guidelines of Insurance Company, NFPA

Water Pollution (Note 1): As per the state No Objection Certification (NOC) and Environmental Clearance (EC) of Project

Air Pollution (Note 1,2,3): As per the state NOC and EC of Project

Noise: Project Standard(Max 85db at distance of 1m from the equipment)

Fire Protection: OISD

Piping Design: ASME B31.3/IBR Material : ASTM/NACE/IS

Concrete: IS

Roads: IRC, Ministry of Surface Transport

Materials: Refer to applicable sections

Mechanical Equipment: Refer to Mechanical section

Welding: ASME

Shell & Tube Heat Exchangers: TEMA / API

Air Cooled Heat Exchangers: API

Tanks: API

Burner Controls: Low NOx (limitation as per expert committee report)

Process Heaters: OISD and respective API's / Project standards

2.4 Design life shall be: As per respective API codes for equipment
For other, Project Standards shall apply

Minimum Design Life shall be :-	
Buildings	25 years
Electrical / Instrumentation	20 years
Structural Steel	25 years
Reinforced Concrete	25 years
Reactors (non-movable trays & Internals)	20 years
Reactors (removable trays & Internals)	20 years
Towers	20 years
Drums	20 years
Air Fin Process Bundle	20 years
Pump Cases	20 years
Turbines	20 years
Heat Exchanger Shells	15 years
Heat Exchanger Channels	15 years
Heat Exchanger Removable Tube Bundles	15 years
High Alloy Exchanger Tubes (> 9% Chromium)	15 years
Piping	15 years
Valves	15 years
Strainers	15 years
Expansion Joints	15 years
Pipelines	30 Years
Replacement Compressor Internals	15 years
Removable Vessel Trays & Internals	15 years
Exchanger Tubes (CS through to 9% Chromium)	15 years
Baffles	15 years
Replaceable Pump Trim & Internals	15 years
Fired Heater Tubing	100,000 h

3 Climatic Data

3.1. Wind

3.1.1 Wind Velocity

Average velocity	
- Summer	37 – 45 km/hr
- Winter	15 – 26 km/hr
Maximum velocity	72 m/sec (259 km/hr)(During 1999 cyclone)
- cyclone	200 - 250 km/hr (S-SE)
Basic wind speed for structural design	65 m/sec (234 km/hr)

A basic wind speed of 65m/s shall be used in accordance with IS 875 (Part 3)-1987 (reaffirmed 2003), Wind Loads.

3.1.2 Wind Direction and Percentage of Time for Each Quadrant

% of time	N	NE	E	SE	S	SW	W	NW	Calm
Morning (0830)	13.8	6.4	2.1	2.4	7.7	21.6	11.0	8.2	(Δ26.8)
Evening (1730)	15.0	6.7	9.41	8.16	15.0	24.8	4.8	1.8	(Δ27.8)

Wind Loading

The applicable standard for wind loading is IS 875, refer also to 3.1.1

Note- The K_2 factor shall be based on a Terrain Category Class 2 and the K_3 factor shall be taken as 1.0.

3.2 Air Temperature

	Temp, °C	Notes
Maximum recorded	42.4	
Minimum recorded	11.3	
Mean dry bulb	30.7	
Mean wet bulb	28.0	
Average max monthly	37	
Average min monthly	16	
Design maximum	42.4	Note 1
Design minimum	11.3	
Design wet bulb	29	

Note 1: Max solar gain temperature on bare piping 65 °C

For Air Coolers:
Design maximum 42°C

3.3 Relative Humidity

RELTIVE HUMIDITY	RH%	Notes
Mean daily maximum	99.7	
Mean daily minimum	24.7	
Average monthly humidity	95% (max) 55% (min)	
Summer mean (months)	90	
Winter mean (months)	45	

3.4 Rainfall

RAINFALL	Value	Notes
Maximum recorded annual	2251.7 mm	
Minimum recorded annual	1018.6 mm	
Average annual	1572.0 mm	
Maximum recorded in 1 hr	125 mm	During 1999 cyclone
Maximum recorded in 24 hrs	335 mm	
Design rainfall (per hour)	125 mm	
Rainy season	May to November	Note 1

Note1: Indian meteorological Department data and the Indian Maritime Authority Data

3.5 Snowfall

Not applicable

3.6 Barometric Pressure

BAROMETRIC PRESSURE	mbar	Notes
Maximum	1010	
Minimum	966.3	
Average	1005	

3.7 Solar Heat

SOLAR HEAT FLUX AT MIDDAY	Kcal/m².hr	Notes
Minimum		
Maximum	678.2	
Average		

3.8 Atmosphere

a) Extreme moisture (tropical climate) YES

b) Marine exposure (salt spray) YES

c) Sand storms NO

d) Copper-attacking fumes (ammonia, sulphur, etc.)
SO_2: 16.6 micro gm (maximum)

e) Exposure to conductive or corrosive dusts (carbon, iron oxide ammonium nitrates or phosphates, etc.)
NO_x: 32.5 micro gm (maximum) at Paradip Phosphate Ltd which is adjoining industry

f) Exposure to corrosive agents (nitric or sulphuric acids, chlorine, caustic, etc.) SO_2 and NO_x as above

g) Exposure to other pollutants originating from surrounding industrial plant NO

3.9 Miscellaneous Site Data

a) Frost Level NA m

b) Thunderstorm Frequency In 30 occasions in 1991

c) Sandstorm Frequency NA

d) Temperature Inversion Occurrence Not Available

e) Ground Temperature. The results of earth temperature measurements are given in the Geotechnical Investigation Report. Readings taken at depths of between 0.6 m and 1.6m were in the range of 29 ^{o}C to 34 ^{o}C

3.9.1 Seismic design shall be as follows;

Seismic design shall be in accordance with IS 1893 Part 1 & 4, and the Earthquake Engineering Analysis Report, prepared by Fugro GeoConsulting/Fugro West, Project No. 3193.026 dated Sept.2008. This report includes site specific seismic response spectra that shall be used to calculate seismic loading. The project site falls under Zone 3 as described in IS 1893.

3.9.2 The maximum recorded flood level is elevation 3910 mm (IMSL)

Remarks: Site shall be raised equal to or above the maximum recorded flood level

4 Economics

4.1 Economic evaluations shall be made as required.

4.2 Additional cost saving proposal or investment for energy efficiency shall have payout period of 3.5 years. (8320 operating hrs/year).

4.3 Taxes, shall not be included in the evaluation.

4.4 Utility costs are-

a.	Steam -	102.0 kg/cm² g:	27.27 $/t
		40.0 kg/cm² g:	23.53 $/t
		14.0 kg/cm² g	20.01 $/t
		4.0 kg/cm² g	15.62 $/t
b.	Water -	Treated Raw Water	0.65 $/t
		Cooling Water	0.04 $/t
		Demineralised Water	1.1 $/t
c.	Condensate - Clean		0.8 $/t
d.	Refinery Fuel Oil		452 $/t
	High Sulphur Fuel Oil		327 $/t
	Fuel Gas		527 $/t
e.	Electrical power (for studies) (tariff may be different)		0.068 $/kWh
f.	Instrument Air		10 $/t
g.	Nitrogen		77 $/t

5 Utilities

All utility values quoted are as provided at the unit battery limit of the consumer, unless otherwise noted. All pressures are referenced to grade.

5.1 Steam

5.1.1 At CPP (or Supplier Unit) Battery Limit

SERVICE	PRESSURE, kg/cm²g				MAX TEMP
	Normal	Max	Min	Design*	°C
HHP steam	104	106	104	121.6	504
HP steam	42	44	40	46	410
MP steam	14.5	15	14	18	305
LP steam	4.5	5	4	7	190
Exhaust					

* at boiler or super-heater outlet

QUALITY REQUIRED	Units	Value
Max TDS	ppm	0.1
Max silica	ppm	0.02
Conductivity	Micro Siemens/cm	0.2

5.1.2 At User Unit Battery Limit

SERVICE	PRESSURE, kg/cm²g				TEMPERATURE, °C			
	Normal	Max	Min	Design	Normal	Max	Min	Design
HHP steam	101	104.0	100	121.6	499	504	471	515
HP steam	40.0	42.0	39	46.0	390	400	380	427
MP steam	14.0	15.0	12.5	18.0	290	305	210	350
LP steam	4.0	5.0	3.3	7.0	175	190	146	240
LLP steam								

Design of ISBL distribution piping shall be such as to meet following minimum pressure at the equipment under maximum flow condition. For steam turbines see next section.

SERVICE	PRESSURE, kg/cm²g
HHP steam	98
HP steam	38
MP steam	12.0
LP steam	3.0

5.1.3 Steam Turbines

The steam conditions listed below apply at the inlet and exhaust flanges of steam turbines.

TURBINE INLET CONDITION

		MINIMUM	NORMAL	MAXIMUM	DESIGN
HHP STEAM	TEMPERATURE,°C	471	499	504	515
	PRESSURE, Kg/cm2(g)	98	99	104	121.6
HP STEAM	TEMPERATURE,°C	380	390	400	427
	PRESSURE, Kg/cm2(g)	38	40	42	46
MP STEAM	TEMPERATURE,°C	210	290	305	350
	PRESSURE, Kg/cm2(g)	12	14	15	18

TURBINE EXHAUST CONDITION

		MINIMUM	DESIGN
HP STEAM	TEMPERATURE,°C	390	427
	PRESSURE, Kg/cm2(g)	42	46
MP STEAM	TEMPERATURE,°C	290	350
	PRESSURE, Kg/cm2(g)	15	18
LP STEAM	TEMPERATURE,°C	175	240
	PRESSURE, Kg/cm2(g)	5	7

Note: Design refers to mechanical design conditions.

Definitions:

Normal	applies to the steam conditions expected in usual operation and at which the highest efficiency of steam turbines is desired. The steam guarantee is usually given at this condition.
Minimum Inlet	refers to the lowest steam pressure and temperature condition at which steam turbines are required to produce the rated horsepower and speed.
Maximum Inlet	refers to the highest steam pressure and temperature condition to which steam turbines may be subjected continuously. (Note1).
Minimum Exhaust	is the lowest exhaust steam pressure at which steam turbines are required to operate continuously.
Maximum Exhaust	is the highest exhaust steam pressure at which steam turbines are required to produce rated horsepower and speed. (Note1).
Rated	refers to the steam condition at which steam turbines are required to develop the rated (nameplate) horsepower and corresponding speed.
Note (1)	The maximum inlet and exhaust pressures and the maximum inlet temperature may exceed the specified values for short periods, as defined by NEMA Standard SM-23 paragraph 2.4.4.14.

5.2 Condensate

5.2.1 Condensate shall be recovered.

5.2.2 Condensate from HP steam system shall discharge at 38.5 kg/cm²g (Normal)

5.2.3 Condensate from MP steam system shall discharge at 12.5 kg/cm²g (Normal)

5.2.4 Condensate from LP steam system shall discharge at 3.3 kg/cm²g (Normal)

SYSTEM CONDITIONS	Pressure kg/cm²g				Temperature °C			
	Min	Max	Normal	Design	Min	Max	Normal	Design
HP	36.5	40.5	38.5	46.0	247	253	250	280
MP	10.5	13.5	12.5	18.0	186	197	194	225
LP	2.3	4.0	3.3	7.0	137	152	147	170
LLP	Atm	0.5	0.3	3.5	105	112	107	140

Condensate will be flashed to LLP steam and pumped to the condensate header at 7.7 kg/cm²g at BL. Small condensate streams will be collected and flashed outside BL and pumped at the same BL pressure.

5.2.5. Clean (Polished) Condensate

Battery limit conditions at CPP

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	4.0	5.5	6.0	9.0
Supply temperature, °C	50	64	70	90

Clean Condensate Quality	Units	Value
Condensate suspended solids	ppm wt	Nil
pH		7.0 – 8.5
Silica as SiO2	ppm wt	0.02 max
Hardness as CaCO3	ppm wt	Nil
Oil Content	ppm wt	Nil
Iron as Fe	ppm wt	0.01 max
Ammonia	ppm wt	0.1 max
Sodium as Na	ppm wt	0.02 max
Dissolved Oxygen	ppm wt	Not measured
Total Dissolved Solids (TDS)	ppm wt	0.5 max
Conductivity at 20°C	Micro S/cm	0.2 max
M alkalinity as CaCO3	ppm wt	Nil
Chlorides	ppm wt	0.02 max

5.3 Water

5.3.1 Cooling Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	4.5	5.5	6.5	8.0
Return pressure, kg/cm²g	2.5	3	3.5	8.0
Supply temperature, °C	-	35	35	65
Return temperature, °C	-	44	44	65

5.3.2 Raw Water

WATER QUALITY	Units	Value
Source	-	Mahanadi River, Cuttak
Availability	MGD	40
Appearance		Clear on settling
Odour		Unobjectionable
pH	-	7.5
Totals dissolved solids	ppm wt	183
Chlorides as Cl	ppm wt	12
Phenolphthalein alkalinity as CaCO3	ppm wt	nil
Turbidity	NTU	500
Total alkalinity	ppm wt	76
Ca hardness as CaCO3	ppm wt	40
Total hardness as CaCO3	ppm wt	73
Magnesium hardness as CaCO3	ppm wt	33
Temporary hardness as CaCO3	ppm wt	73
Permanent hardness as CaCO3	ppm wt	nil
Dissolved silica as SiO2	ppm wt	4
Colloidal silica	ppm wt	2 (Note 1)
Dissolved iron as Fe	ppm wt	< 0.01
Sulphates as SO4	ppm wt	< 2.0
Carbonates as CaCO3	ppm wt	Nil
Bicarbonates as CaCO3	ppm wt	76
Magnesium as Mg	ppm wt	7.9
Sodium as Na	ppm wt	12
Phosphate as P	ppm wt	< 0.05
Nitrates as N	ppm wt	0.5

Note 1: Occasional seasonal excursion in Colloidal Silica up-to 5 mg/l is expected

5.3.3 Treated Raw Water

WATER QUALITY	Units	Value
pH		7-8.5
Colour	Hazen	5.0
Smell		Agreeable
Taste & Odour		Unobjectionable
Turbidity	NTU	1.0 max
Totals dissolved solids	ppm wt	150 max
Total Hardness	ppm wt	85 max
Chlorides (as Cl)	ppm wt	15 max
Sulphate (SO4)	ppm wt	60 max
Total Iron (Fe)	ppm wt	0.01 max
Dissolved Silica	ppm wt	4 max
Colloidal Silica	ppm wt	2 max (Note 1)

Note 1: Occasional seasonal excursion in Colloidal Silica up-to 5 mg/l is expected

5.3.4 RO Permeate Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	-	8.0	-	10.0
Supply temperature, °C	-	Ambient	-	65

WATER QUALITY	Units	Value
pH	-	5.7-6.7
Total Hardness as CaCO3	ppm wt	6
Total dissolved solids	ppm wt	100 max
Alkalinity as CaCO3	ppm wt	20
Chlorides as Cl	ppm wt	40
Sulphate as SO4	ppm wt	5.0
Silica as SiO2	ppm wt	0.5 max
Colloidal Silica	ppm wt	0.01 max
TOC	ppm wt	0.2 max

5.3.5 Demineralised Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	4.0	5.5	6.0	10.0
Supply temperature, °C	-	Ambient	Ambient	65

WATER QUALITY	Units	Value
pH	-	6.5 – 7.5
Hardness	ppm wt	Zero
Total dissolved solids	ppm wt	0.1 max
Conductivity at 20 °C	micromho/cm	0.2 max
M alkalinity as CaCO3	ppm wt	Nil
Chlorides	ppm wt	Nil
Iron as Fe	ppm wt	0.01 max
Silica as SiO2 (Note 1)	ppm wt	0.02 max
Colloidal Silica	ppm wt	0.01 max
TOC	ppm wt	0.2 max
Sodium as Na	ppm wt	0.1 max

Note 1: Include Colloidal Silica

5.3.6 Boiler Feed Water

CONDITIONS	Min	Normal	Max	Mech Design
HP supply pressure, kg/cm²g	58.0	58.0	60.0	82.0
HP supply temperature, °C	105	105	116	150
MP supply pressure, kg/cm²g	28.0	28.0	30.0	40.0
MP supply temperature, °C	105	105	116	150

WATER QUALITY	Units	Value
pH	-	8.5 – 9.5
Conductivity at 20 °C	micromho/cm	0.2 max
Chlorides	ppm wt	1.0 max
Iron as Fe	ppm wt	0.01 max
Silica as SiO2 (Note 1)	ppm wt	0.02 max
Colloidal Silica	pm wt	0.01 max
Oil Content	ppm wt	Nil
Oxygen	ppm wt	0.005 max

Note 1: Include Colloidal Silica

5.3.7 Service Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	4.0	6.0	8.0	10.5
Supply temperature, °C	ambient	ambient	ambient	65

5.3.8 Drinking Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g*	4.0	6.0	6.5	12.5
Supply temperature, °C	Ambient	Ambient	Ambient	65

*supply conditions for the refinery units

5.3.9 Fire Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	7.0	7.0	11.5	16.0
Supply temperature, °C	Ambient	Ambient	Ambient	65

Conditions are at furthest hydrant.

5.3.10 Stripped Sour Water

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure, kg/cm²g	3.0	5.0	6.0	10.0
Supply temperature, °C	20	40	50	65

5.4 Air

5.4.1 New plant and instrument air systems shall be provided

5.4.2 Plant Air

CONDITIONS	Min	Normal	Max	Mech Design
Header pressure, kg/cm²g	4.0	7.0	8.0	10.5
Supply temperature, °C	40	40	50	65

QUALITY	Value
Design Dew point	Ambient
Oil content	Nil (oil free)

5.4.3 Instrument Air

CONDITIONS	Min	Normal	Max	Mech Design
Header pressure, kg/cm²g	4.5	7.0	8.0	10.5
Supply temperature, °C	ambient	40	50	65

QUALITY	Value
Design Dew point	-40°C at Atmospheric Pressure
Oil content	Nil (oil free)

Remarks: Electrical Compressor driver

5.5 Nitrogen

5.5.1 Nitrogen generation facility shall be provided based on air liquefaction.

5.5.2 Nitrogen conditions

CONDITIONS	Min	Normal	Max	Mech Design
Header pressure, kg/cm²g	3.0	8.0	8.3	10.5
Supply temperature, °C	20	amb	amb	65

QUALITY	Value
Nitrogen purity	99.99% v (min)
Dew point	-80 °C at atmospheric pressure
Oxygen content	10 ppm vol max.
Oil content	Nil

Note: Dedicated Nitrogen header for VGO Hydrotreater unit is provided, supplying Nitrogen at 80.0 kg/cm^{2}g to this unit.

5.6 Fuels

5.6.1 Refinery Fuel Gas

Fuel gas from the refinery collection and distribution network shall be used as a fuel on the following basis:

CONDITIONS	Min	Normal	Max	Mech Design
Supply pressure (consumer), kg/cm²g	3.2	3.9	4.2	6.0
Supply temperature (consumer), °C	Ambient	40	55	65
Supply pressure (producer), kg.cm^{2}g	4.5	5.0	5.0	6.0
Supply temperature (producer), °C	Ambient	40	50	65

QUALITY	Units	Value		
FUEL TYPE		Full Refinery	Fuels Refinery	Fuel Gas
Net calorific value (LHV)	kcal/kg	11,070	11716	_____
Molecular weight	-	20.1	21.1	_____
Composition				
H2S	vol%	0.005	-	0.005
H2	vol%	-	13.15	23.70
CO	vol%	4.0	-	0.30
N2	vol%	0.7	-	5.70
O2	vol%	-	-	0.20
C1	vol%	80.48	38.95	30.30
C2=	vol%	-	31.99	16.9
C2	vol%	7.7	13.64	15.80
C3=	vol%	-	-	3.5
C3	vol%	4.48	1.29	0.60
C4=	vol%	-	-	0.1

C4	vol%	2.4	0.98	1.20
C5+	vol%	0.22	-	1.20
C6+	vol%	0.02	-	-
H2O	vol%	-	-	0.6

5.6.2 Refinery Fuel Oil

FUEL CONDITIONS	Units	Minimum	Normal	Maximum	Design
Supply header pressure	kg/cm²g	8	10.0	11.0	15.0
Pressure available at burner	kg/cm²g	5.5	8	11.0	15.0
Return header pressure	kg/cm²g	2.0	2.5	3.0	15.0
Temperature at burner	°C	Ambient	40	40	65.0

FUEL QUALITY	Units	Minimum	Normal	Maximum	Design
Specific gravity	SG		0.962		
Lower heating value (LHV)	kcal/kg		9480		
Viscosity @ 40 °C	cP		8.7		
Pour point	°C		8 max		
Flash point	°C		>66		
Sulphur content	ppmw		2500 max		
Vanadium content	ppm wt		2 max		
Nickel content	ppm wt		1 max		
Iron content	ppm wt		0.5 max		
Nitrogen content	ppm wt		60 max		
AROMATICS CONTENT					
Diethyl Benzene	ppmw		110		
C_{11}+ Aromatics	wt%		5.38		
Tetramethyl	wt%		1.29		
C/H ratio			8.8		

5.6.3 High Sulphur Fuel Oil (for Utility Boilers)

HSFO blend components & composition

Stream	Source	Blend % Range	Typical Blend	
Light Cycle Oil (LCO)	FCC	0-37	37	-
Clarified Oil (CLO)	FCC	25	25	-
Light Coker Gas Oil (LCGO)	DCU	27	-	27
Atmospheric Gas Oil (AGO)	AVU	0-35	-	35
Vacuum Residue (VR)	AVU	38-54 (Note 1)	38	38
Heavy Aromatics	Future Full Refinery			

Note 1: Higher % of VR is possible with Kuwait Crude Processing.

HSFO CONDITIONS	Units	Minimum	Normal	Maximum	Design
Supply header pressure	kg/cm²g	3.0	3.0	5.0	15.0
Supply Temperature	°C	70	70	90	115

HSFO QUALITY	Units	Minimum	Normal	Maximum	Design
Specific gravity	SG	0.93	1.0	1.01	1.03
Higher heating value (HHV)	kcal/kg				
Lower heating value (LHV)	kcal/kg	9536	9624	9804	9310

Viscosity @ 70 °C	cP	12.5	27.5	225.7	Note 1
Viscosity @ 130 °C	cP	3.1	5.4	18.5	Note 1
Pour point	°C	26	32	35	62
Flash point	°C	70	70	89	62
Sulphur content	Wt%	2.9	2.9	3.8	4.9
Vanadium content	ppm wt	77	90	90	100
Nickel content	ppm wt	15	15	27	66
Nitrogen content	ppm wt	2200	2200	2900	5010
Silica content	ppm wt		1		
C/H ratio		7.9	9	9.2	9.5
CCR	wt%	11	11	14	14
Asphaltene	wt%	5	5	7	7
Note 1: Viscosity in all cases is below design viscosity, which is 20cSt at 180°C					

5.6.4 Gas Turbine Fuel Oil

Gas Turbines Fuel Oil is HDT Diesel/Atmospheric Gas oil (SR Diesel) or HDT Naphtha/SR Naphtha. The fuel is pumped from GT Diesel tanks, 110-TK-004/005, or GT Naphtha Tanks, 110-TK-001/002 (these tanks are located close to but outside CPP battery limit) to Diesel or Naphtha Day Tanks respectively inside CPP battery limits. The conditions below refer to supply conditions at CPP battery limit.

FUEL CONDITIONS	Units	Minimum	Normal	Maximum	Design
Supply header pressure	kg/cm²g	3	3	4	8.0
Supply Temperature	°C		40		65

QUALITY OF DIESEL AS GT FUEL OIL					
	Units	Minimum	Normal	Maximum	Design
Specific gravity	SG		0.82-0.85		
Higher heating value (HHV)	kcal/kg		10748		
Lower heating value (LHV)	kcal/kg		10032		
Viscosity @ 40 °C	cP		1.0-2.5		
Pour Point	°C				
Flash Point	°C				
Sulphur content	ppm wt		500 max		
QUALITY OF TYPICAL ALTERNATIVE GT FUEL OIL (Naphtha)					
	Units	Minimum	Normal	Maximum	Design
Specific gravity	SG		0.78		
Higher heating value (HHV)	kcal/kg		11159		
Lower heating value (LHV)	kcal/kg		10414		
Viscosity @ 40 °C	cP		0.87		
Vapour Pressure	kg/cm^2a		0.8		
Sulphur content	ppm wt		0.50		
Iron content	ppm wt		0.10		
Nitrogen content	ppm wt		0.50		

5.7 Hydrogen

5.7.1 A new hydrogen gas generation and distribution system shall be provided.

5.7.2 Hydrogen conditions

CONDITIONS	Min	Normal	Max	Mech Design
Header pressure, kg/cm²g		25.0		28
Supply temperature, °C		45		70

QUALITY	Value
Hydrogen purity	99.9% v (min)
CO + CO2	10 ppmv (max)
N2 + CH4	Balance

5.7.3 Feed Stock Naphtha quality - Normal Feedstock

Sl. No	Attribute	BOO option Value	FEL phase Value	Remarks
	Specific gravity	0.670	0.651	
	Total Sulphur	1.0 ppmw max	0.5 ppmw	
	Total Nitrogen	1.0 ppmw max	0.5 ppmw	
	Total Chloride	1.0 ppmw max	1 ppmw	
	Heavy metals	100 ppbw max	100 ppbw As+Pb+Ni+V+Cu	
	ASTM Distillation	Deg C	-	
	IBP	48.8		
	10%	50.7		
	30%	65.6		
	50%	68.2		
	70%	71.2		
	90%	76.2		
	FBP	89.7		
	C/H ratio	5.66	-	
	LHV	10,628 Kcal/kg	-	
	PONA	84.7 / 0.0 / 13.0 / 2.3 wt%	-	
	Water content	100 ppmw	-	
	Composition	Nil	Wt%	
	n-Butane		0.02	
	i-Pentane		17.28	
	n-Pentane		33.36	
	C5-olefin		0.01	
	Cyclopentane		2.57	
	2,2-Methylbutane		0.5	
	2,3-Methylbutane		1.97	
	2-Methylpentane		13.91	
	3-Methylpentane		9.41	
	n-Hexane		16.68	
	Methylcyclopentane		2.32	
	Cyclohexane		0.21	
	Benzene		1.39	
	2,2-Dimethylpentane		0.02	
	2,3-Dimethylpentane		0.01	
	2,4-Dimethylpentane		0.05	
	2-Methylhexane		0.03	

	3-Methylhexane		0.02	
	Dimethylcyclopentane		0.03	

Alternative feedstock - C3-C4 mix:

Sl.No	Attribute	BOO option Value	FELphase Value	Remarks
1.	Composition	Vol % Propane -21.9 Butane - 78.1	Wt% Ethane -0.3 Propane-18.6 i-Butane-16.2 n-Butane-63.9 C5+ -1.0	Alternative feed in FEL phase was Sulfolane Raffinate
2.	Total Sulphur	200 ppmw max	8 ppmw	
3.	Water content	100 ppm max	-	

5.7.4 Fuel Gas Quality

Sl.No	Attribute	BOO option Value	FELphase Value	Remarks
1.	Net calorific value (LHV) (Kcal/kg)	To be determined by BOO operator	11,070	
2	Molecular weight	20.1	20.1	
3	H2S content	50 ppm vol	50 ppm vol	
4	Composition	mole % H2 - 23.7 CH4 - 30.3 C2H6-15.8 C2H4 - 16.9 C3H6- 3.5 C3H8 -0.6 n-C4 - 0.8 i- C4 - 0.4 C5+ - 1.7 H2O - 0.6 N2 - 5.7	vol% H2 - - CO - 4.0 N2 - 0.7 C1 - 80.48 C2 - 7.7 C3 - 4.48 C4 - 2.4 C5 - 0.22 C6+ - 0.02	

5.7.5 HP Steam Quality

QUALITY REQUIRED	Units	Value - Max.
Max TDS	ppm	0.1
Max silica	ppm	0.01
Conductivity	Micro Siemens/cm	0.2
Methanol content -	ppmw	20

5.8 Coke

Petcoke can be considered as fuel for CFBB/ IGCC for steam/ power generation

5.9 Electric Power

Refer to Section 12 for details of electrical supply.

6 Pressure Vessels and Tankage

6.1 Welded Unfired Pressure Vessels
For Vessels with a Design pressure greater than 1 barg the following shall apply

6.1.1 Codes and Standards

a) Project Standard PDRP-8430-SP-0001 General Requirements for Welded Unfired Pressure Vessels.

b) Project Standard PDRP-8440-SP-0010 General Requirements for Welding

6.1.2 The following International and National Codes shall apply:

Design:	ASME SEC VIII, DIV 1 / 2, IBR, OISD
Materials:	ASTM

6.1.3 There shall be no conflict in Codes and Standards. The most stringent requirements of all of the above shall apply.

6.2 Tankage
For Atmospheric and Low Pressure Tanks the following shall apply

6.2.1 Codes and Standards
a) Project Standard PDRP-8430-SP-0002 API Storage Tanks

b) Project Standard PDRP-8440-SP-0010 General Requirements for Welding

6.2.2 The following International and Local Codes shall apply:

API Standard 650	Welded Steel Tanks for Oil Storage
API Standard 620	Recommended Rules for Design and Construction of Large, Low Pressure Storage Tanks
API RP 2000	Venting Atmospheric and Low Pressure Storage Tanks
Materials:	ASTM/IS

6.2.3 Silos
Silos are normally designed to various codes, fabrication/weld details and inspection requirements.
Typical Standards for Silo design are, DIN 1055 Part 6 and ISO TC 98/SC3/WG5.
However, the design (including the allowable stresses) shall meet the requirements of API 650 as a minimum.

6.2.4 There shall be no conflict in Codes and Standards. The most stringent requirements of all of the above shall apply.

7 Heat Transfer Equipment

There shall be no conflict in the Codes and Standards. The most stringent requirements shall apply.

7.1 Shell and Tube Exchangers -

7.1.1 Shell and Tube exchangers shall be designed in accordance with:

Project Standard PDRP-8430-SP-0003 General Requirements for Shell and Tube Heat Exchangers

7.1.2 The following International codes shall apply:

TEMA R	ASME SEC VIII DIV 1

7.1.3 The preferred tube size for carbon and low alloy steel shall be 25mm diameter, 2.5mm thick, 6000 mm straight length.

7.1.4 Tube diameters and thicknesses in accordance with TEMA Table RCB 2.21 are permitted.

7.1.5 The maximum straight length for tubes in removable bundles shall be 7315 mm (24 feet). Tube lengths for non-removable bundles may exceed 7315mm, subject to IOCL approval on a case by case basis, but shall not exceed 12192mm. Total straight tube lengths on all removable bundle exchangers shall be limited to a maximum of four different lengths for standardisation purposes – 4267 mm,4877 mm, 6000 mm and 7315 mm..

7.1.6 The maximum shell diameter shall be 2540 mm (100 inches).

7.1.7 The maximum bundle diameter for removable bundle exchanges shall be 1524 mm (60 inches).

7.1.8 The maximum removable bundle weight shall be 20,000 kg (44,092 lb).

7.1.9 The ratio of tube length to bundle outer diameter for removable bundles shall be less than 10:1.

7.1.10 The maximum tube length for vertical thermosyphon reboilers shall be 6096 mm (20 ft).

7.1.11 IOCL will advise a list of equipment items where the material selection for heat exchanger tubes shall be upgraded to admiralty brass as a minimum. In cases where admiralty brass is considered unacceptable, IOCL approval of the selected material shall be obtained. IOCL approval shall be obtained where revision to design is required as a result of the material change.

7.2 Double Pipe Exchangers

7.2.1 Double Pipe exchangers shall be designed in accordance with:

Project Standard PDRP-8430-SP-0005 General Requirements for Double Pipe and Multiple Exchangers

7.2.2 The following International code shall apply:

ASME SEC VIII,DIV 1

7.2.3 The maximum tube length shall be determined on a case by case basis, following the requirements of the applicable process datasheet.

7.3 Air Coolers

7.3.1 Air coolers shall be used to the following extent:

Air cooling shall be maximised for which the cut off temp of the process stream shall be 55ºC. However, when further trim cooling by water is necessary, the cut off temp of stream shall be 65ºC. However, to avoid installation of small trim cooler, the guideline can be relaxed.

7.3.2 Air cooled exchangers shall be in accordance with:

Project Standard PDRP-8430-SP-0004 General Requirements for Air Cooled Heat Exchangers

Remarks: None

7.3.3 The following International Codes shall apply:

API 661

7.3.4 The design air inlet temperature shall be
_____ °C dry bulb. Refer to section 3.2

7.3.5 The minimum ambient temperature shall be
_____ °C dry bulb. Refer to section 3.2

7.3.6 Fans (forced or induced) shall be as specified in 7.3.2 above (Forced draft fans are preferred).

7.3.7 The maximum tube lengths for Air Coolers shall be12.200 m

7.3.8 Winterising required No

7.3.9 Vibration switches required Yes

7.4 Waste Heat Boilers

7.4.1 Waste heat boilers generating steam shall be designed in accordance with:

Indian Boiler Regulations 1950 (IBR)

7.5 Fired Heaters

7.5.1 Fired Heaters shall be designed in accordance with:-

API Standard 560, API Standard 530, OISD requirements

Remarks:
For the BFW Economiser, Steam Generator, and Steam Superheat Convection Section services Indian Boiler Regulations 1950 (IBR) shall apply.

7.5.2 The following National and Local codes shall apply:

Indian Boiler Regulations 1950 (IBR)

7.5.3 Heaters shall be equipped with (combination) burners, except for applications in the Hydrogen, Coker and CCR units.

7.5.4 Permanent pilot burners shall be provided for each burner.

7.5.5 The minimum efficiency to be considered is 92 %

7.5.6 The minimum stack height required above grade line is detailed in Section 2.4 of Environmental Design Basis, PDRP4280-8150-PH-0002.

7.5.7 The stack type shall be defined by datasheet

7.5.8 Air heaters shall be considered on a case by case basis.

7.5.9 Soot blowers
Provision for soot blowers shall be provided for liquid fuel firing only.

8 Mechanical Equipment

8.1 Pumps, Compressors and Turbines

8.1.1 Pumps, compressors and turbines shall be specified in accordance with Project standards for rotating equipment.

Remarks: None

8.1.2 The following International Codes shall apply:
Generally : API for respective equipment. Others to be identified and included as required

8.1.3 Direct acting steam reciprocating pumps shall not be acceptable for limited use.

8.1.4 Air blower design shall use 90 % relative humidity

42.4 °C dry bulb temperature.

Refer to section 3 for additional climatic data.

8.1.5 The steam conditions for steam turbines are given in Section 5 Utilities.

9 Civil Engineering

9.1 Site Preparation

State of Site (cleared, levelled etc.) South side of creek: rough graded, including underground services, some buildings and vegetation. North side of creek: land reclamation works in progress.

Existing site levels are shown on Topographic Drawing No: IOCL/FGTL/PD/SURVEY/PRODUCT_CRUDE/DWG/01A by Fugro Geotech Ltd.

The Topographical Survey drawing of the site is available from IOCL upon request.

9.2 Ground Conditions

9.2.1 Ground conditions input is provided in the following geotechnical reports:

i) Paradip Refinery Project -Geotechnical Engineering Consultancy, Final Report -Part 1, Paradip , Orissa State India, Fugro Report No. C696-04, Issue 01, Nov 2008

ii) Final Report, Earthquake Engineering Analyses, Paradip Refinery Project, Orissa, India, Fugro Project No. 3193.026, Sept 2008

iii) Paradip Refinery Project -Geotechnical Engineering Consultancy, Final Report -Part 3, Paradip , Orissa State India, Fugro Report No. C696-04, Issue 01, Nov 2008

iv) Draft Interpretation Report on Geotechnical Investigation for Paradip Refinery Project at Paradip, Orissa, India, Fugro Report no 213/08/R-0, Nov 2008

v) Preliminary Factual Report on Geotechnical Investigation for Paradip Refinery Project at Paradip, Orissa, India, Fugro Report no 213/08/R-0, Aug 2008.

IOCL consultant is Fugro Geotech Limited

9.2.2 Underground obstructions include existing underground pipelines and cables.

9.2.3 Foundations:

Foundations design shall be in accordance with the following Project Specifications

PDRP-8310-SP-0004	Design of Blast Resistant Buildings for High Hazard Plants
PDRP-8310-SP-0006	General Specification for Piling
PDRP-8310-SP-0007	Design of Concrete Foundations Structures & Paving
PDRP-8310-SP-0008	Design of Storage Tank Foundations
PDRP-8310-SP-0009	Design of Foundations and Structures Supporting

Machinery

Applicable IS standards which apply shall govern in case of conflict with project specification

a. Information on the ground water level is provided in the Geotechnical Reports.

b. Allowable net bearing value varies across the site and shall be determined on the basis of Geotechnical Reports and proposed development.

c. Design Frost line depth is NA m.

d. Piling requirements to be determined on the basis of specific design conditions.

e. Ground improvement is required.

9.2.4 Piles
- type & size — As per design requirement
- capacity — As per design requirement
- maximum uplift — As per design requirement
- lateral resistance — As per design requirement

9.2.5 Type of foundations:

a) Major equipment, buildings and structures Spread footings & piled foundations, as required / recommended in the Geotechnical Reports and meeting design criteria shown in the project specifications.

b) Minor equipment, pipe supports Spread footings

c) Possibly Stone columns for storage tank foundations or other suitable for purpose type of ground improvement.

9.3 Water Run-Off/Sewers

9.3.1 a) Paving shall be provided in process area.

Refer to project specification PDRP-8310-SP-0013 and typical details PDRP-8310-45-0009.

b) Access ways shall be paved.

c) Run-off design factors - Paved areas FW = 1.0 :As per standard

Unpaved areas Refer to PDRP-8310-SP-0018

d) Roadways shall be paved to the following standards:

PDRP-8310-SP-0013 Specification for Roads and Paving

9.3.2 Surface drainage systems shall be designed in accordance with:

PDRP-8310-SP-0018 Drainage Basic Design Specification

Remarks: Refer to rainfall data SECTION 3.4

9.3.3 The following drainage systems shall be provided:

Clean water sewer YES

Oily water sewer Accidentally Oil Contaminated (AOC) & Continuously Oil Contaminated (COC)

Sanitary sewer YES

Chemical sewer YES

Other Spent Caustic

Remarks: Spent caustic from all units shall be collected at a common place, from where it will be sent to ETP after due neutralisation. Separate amine sump shall be provided for individual Amine Regeneration Units

9.3.4 Additional sloping of grade shall be provided in fire risk areas, on a case by case basis if specifically identified following HAZOP reviews.
Remarks: None

9.3.5 Sewer pipe shall be designed in accordance with:

IS Standard practices and OISD requirements

Remarks: Manholes shall have water seals

9.4 Waste Treatment Facilities

9.4.1 New waste treatment facilities are to be provided for Oily, Sanitary, Chemical and Spent caustic waste streams:

An incinerator for solid/sludge waste is to be provided.

Land fill facilities are required for non-hazardous waste materials.

9.4.2 The following National and Local Codes shall apply:

Proposed Effluent and Emission Standards for Petroleum Oil Refineries

Issued by the Central Pollution Control Board (CPCB)

9.4.3 Permit application for waste disposal shall be obtained by IOCL

Remarks: None

9.4.4 Permits shall be obtained by IOCL.

Remarks: NOC under waste and Air acts and Environment Clearance shall be obtained by IOCL

9.5 Unit Elevation

9.5.1 Final plant elevation has been established as +4.360m (IMSL) Units & Utilities and +4.060m Tankage & Other Refinery Areas, which shall be considered as high point of paving.

9.5.2 Minimum height for top of foundations above high point of paving shall be per PDRP-8310-SP-0007, Design of Concrete Foundations Structures & Paving.

9.6 Structural Materials

9.6.1 a) Structural material (shall) be in accordance with.

PDRP-8310-SP-0005 General Specification for Concrete Works
PDRP-8320-SP-0017 Structural Steelwork Fabrication

IS shall be followed

b) Additional special structural materials and locations where required are as follows: None

9.6.2 For specification of elevated walkways and platforms see PDRP-8320-SP-0017.

9.7 Buildings

9.7.1 Buildings shall be in accordance with Project, IS, NBC, OISD standards and PDRP-8330-SP-0020

9.7.2 The following National and Local building codes shall apply:

NBC / BIS / OISD

9.7.3 a) Air conditioning shall be provided for the Central Control Room (CCR).

b) Other buildings where air conditioning is required are:
All buildings continuously manned by personnel or containing permanent equipment which requires a controlled environment. See PDRP-8340-SP-0021

9.7.4 a) Other buildings where pressurising is required are:

All Control Rooms
All Substations & Satellite Rack Room (SRR)

9.7.5 Buildings where ventilation and heating are required are:-

All buildings that are occupied or house sensitive electrical equipment shall be ventilated and/or air conditioned. No heating required. See PDRP-8340-SP-0021

9.7.6 Winter design ambient for heating is 12 °C dry bulb.

9.7.7 Summer design ambient for air conditioning is:

39 °C dry bulb.

30 °C wet bulb.

9.7.8 Inside building design temperature is _____ °C.

Remarks: Refer to HVAC Performance Specification. PDRP-8340-SP-0021

9.8 Security

9.8.1 Perimeter fencing shall be provided.

Type: Boundary wall

9.8.2 Area fencing shall be provided.

Type: Refer Fencing Details, PDRP-8310-45-0010

Areas where fencing is required: Refer to Plot Plans

9.9 Roads

	Primary Roads	Secondary Roads	Service Roads
Minimum width (m)	12	7.5	4.0

Refer to
PDRP-8310-SP-0013 - Specification for Roads and Paving
PDRP-8310-45-0006 - Details of Roadways

10 Piping

10.1 Piping shall be designed in accordance with the latest edition plus Addenda of the ASME Code for Pressure Piping: Chemical Plant and Petroleum Refinery Piping, ASME B31.3; Power Piping, ASME B31.1; and other sections which apply as listed below in remarks:

10.1.1 Piping shall, be designed in accordance with PDRP-8230-SP-0002 and PDRP-8230-SP-0004.

Remarks: OISD/IBR/NACE shall be followed as applicable

10.1.2 Category D and Category M Fluid Services

ASME/ANSI B31.3 requires IOCL to identify those fluid services which are in Category D and Category M.

a) Category D - A fluid service in which all the following apply:

(1) The fluid is non-flammable and nontoxic.
(2) The design pressure does not exceed 150 psig. (10.55 kg/cm²ga)
(3) The design temperature is from -20°F (-29°C) to 366°F (186°C)

These services are as follows: As per process requirements

b) Category M - A fluid service in which a single exposure to a very small quantity of a toxic fluid, caused by leakage, can produce serious irreversible harm to persons on breathing or bodily contact, even when prompt restorative measures are taken.

These services are as follows: None

10.1.3 Severe Cyclic Service Lines

ASME/ANSI B31.3 requires the designer to identify those lines subject to severe cyclic conditions. These conditions exist where S_E exceeds 0.8 S_A and the equivalent number of cycles exceeds 7000; or where other conditions will produce an equivalent effect.

These lines are as follows: Categories of fluid and cyclic service

Conditions shall be per process/utilities requirement

11 Instrumentation

11.1 Instrumentation and Control Systems will be furnished in accordance with Engineering Flow Diagrams and the applicable Instrument standards

- PDRP-8550-SP-0001 Instrumentation for General Hydrocarbon and Utility Services
- PDRP-8120-PH-0001 Process Control & Operating Philosophy

12 Electrical

12.1 General requirements for electrical equipment, materials and design shall be in accordance with:

[x] Project Standard PDRP-8531-SP-0001 Electrical Standard

[x] Other IS, IEC

Remarks: None

12.2 Power Supply

The system characteristics shall be as follows:

12.2.1 Incoming supply: 11kV Volts 3 Ph. 50 Hz 10 MVA.

12.2.2 In-plant generation: 11k Volts 3 Ph. 50 Hz 362 MW.
In Plant Distribution Voltage: 66 kV, 3 Ph, 50 Hz

12.2.3 The power source provided by CPP (Captive Power Plan)

12.2.4 The maximum 3 phase fault level is 3600 MVA at 66 kV. Based on max. S/C board rating of 31-5kA @ 66kV

12.2.5 An emergency power distribution shall be provided by Main Electrical Contractor (MEC) to supply process critical loads and emergency lighting, as advised by the Unit Contractor

12.2.6 An un-interruptible power supply shall be provided by the Power Distribution Contractor

Remarks: For DCS PLC and Instrumentation - As advised by Unit Contractor(s).

For HV Switchgear closing / tapping - (determined by Power Distribution Contractor).

For Telecommunications requirements - As advised by Main Telecom Contractor

12.3 Utilisation Voltages

Service	Power Range kW	Voltage Volts	Phase	Freq Hz	Notes
Electric Motors	4 MW	11kV ±6%(AC)	3	50±3%	To be confirmed by System Study.
Electric Motors	>160 ≤4 MW	6600±6% (AC)	3	50±3%	To be confirmed by System Study.
Electric Motors	0.18 - 160	415±6% (AC)	3	50±3%	
Electric Motors	< 0.18	240±6% (AC)	1	50±3%	

Lighting& Small Power	-	240±6% (AC)	1	50±3%	
Critical Lighting		220V (DC)			Through centralised Battery System
Instruments	-	110 (AC)	-	-	Through Dedicated UPS
UPS	-	110 (AC)			
Switchgear Control		110 (DC)			Through Substation DC System
Telecomm/IT Security Systems/CCTV		240±6% (AC			Through Dedicated UPS

12.4 Distribution and Lighting

12.4.1 Service

		Under Ground	Above Ground	Direct Buried	Conduit	Tray
Power	- ISBL	[x]	[x]	[x]	[ ]	[x]
	- OSBL	[x]	[x]	[x]	[ ]	[x]
Lighting	- ISBL	[x]	[x]	[x]	[x]	[x]
	- OSBL	[x]	[x]	[x]	[ ]	[x]
Communications	- ISBL	[x]	[x]	[ ]	[x]	[x]
	- OSBL	[x]	[x]	[ ]	[x]	[x]
Other: specify _____		[ ]	[ ]	[ ]	[ ]	[ ]

12.4.2 Cable Types

Refer to Power & Control Cables: PDRP-8531-SP-0012

12.4.3 Area Classification (shall) be designated by applying the following – IS 5572 and PDRP-8531-SP-0001: Electrical Standard

12.4.3 One-Line Diagrams (OLDs) shall be provided by MEC for site distribution and switchgear / MCC's and by each Unit Contractor for field lightening and small power.

Remarks: For Field mounted Lighting and small panel boards only, MEC is responsible for ALL OLD's associated with equipment located within the substations.

12.4.4 The location of substations and motor control stations shall be selected by each Unit Contractor, based on IOCL preliminary design.

Remarks: MEC is responsible for determining quantities and location of Substations. Unit Contractor is responsible for location of Motor Local Control Stations, and other field mounted equipment such as Panel boards.

12.4.6 Lighting fixtures shall be:

Type	Areas or Locations
High Efficiency Incandescent	As per operational requirement
High Efficiency Fluorescent	As per operational requirement
High Efficiency Mercury Vapour	As per operational requirement

Other None

12.4.7 Underground Cable Installations (Refer also to Section 3.9 Miscellaneous Site Data, e) Ground Temperature)

Soil temperature (summer average) 32 °C
Soil thermal resistivity 1.5 m°C/W

12.4.8 Supplementary Lighting shall be provided

Emergency All areas

Aviation obstruction All Chimneys / Stacks larger than 40m

Security All watch towers

Other General Area Lightening provided by MEC

12.4.9 Offsite lighting shall be provided for Street, Perimeter, Equipment to supplement General Area lightening as required.

12.5 Miscellaneous Requirements

12.5.1 A fire alarm system shall be provided.
(Also see Section 14 Safety Facilities)

12.5.2 A telephone system shall be provided (voice over IP).

12.5.3 A trunked radio communications system shall be provided.

12.5.4 A video conferencing system shall be provided (LAD) and wireless LAN.

12.5.5 Local Area Network shall be provided.

12.5.6 A process CCTV system shall be provided.

12.5.7 An inplant paging system shall be provided.

12.5.8 An intrusion direction and assessment system shall be provided.

12.5.9 A Site access control system (SACS) shall be provided.

12.5.10 Systems shall be on a structured cabling system including outside plant (OSP) areas.

Remarks: Scopes of equipment supply are to be determined, as the majority of the control and power cabinets associated with telecoms will reside within the Power Distribution Contractors Substations. MTC is responsible for provision of all telecoms/systems. Data System highways provided by EPCm-1.

13 Protective Cover

13.1 Insulation for Hot Piping and Equipment

13.1.1 Insulation for hot piping and equipment shall be in accordance with Project Standard PDRP-8440-SP-0004 Hot Service Insulation

Remarks: None

13.2 Insulation for Cold Piping and Equipment

13.2.1 Insulation for cold piping and equipment shall be in accordance with Project Standard PDRP-8440-SP-0005 Cold Service Insulation

Remarks: None

13.3 Winterising

Not applicable

13.4 External Heating

13.4.1 External heating for piping and equipment shall be by:
[x] steam tracing; [x] steam jacketing; [x] electric tracing; [x] other

Remarks: (As per process requirements)

13.5 Personnel Protection

13.5.1 Piping and equipment operating at or above 60 ^{o}C shall be provided with insulation guards for personnel protection where accessible.

Remarks: Above 200°C insulation may be used when approved by IOCL.

13.6 Fireproofing

13.6.1 Fireproofing of equipment and supports shall be provided in accordance with: OISD guidelines and PDRP-8440-SP-0007 Passive fire protection & external fireproofing of equipment support and structure.

Remarks: None

13.7 Painting

13.7.1 Painting for all piping, equipment structures and all externals shall be in accordance with Project Standard PDRP-8440-SP-0006 Protective Paint and Coatings

Remarks: As per process requirements. Includes paint colour coding. (linings of internals are excluded).

13.8 Coating Underground Pipe

13.8.1 Coating of underground steel pipe shall be in accordance with Project Standard PDRP-8440-SP-0008 Coating of Buried Pipework

Remarks: None

13.9 Galvanising

Galvanising of iron and steel items shall be accordance with ASTM A153.

Remarks: Hot dip galvanised gratings. Suitable Piping material

to be procured from vendors

14 **Safety Facilities** (not covered elsewhere)

14.1 Fire Fighting

Fire Fighting facilities shall be provided in accordance with Fire Detection and Protection Specification PDRP4240-8140-SP-1001

Remarks: OISD Stipulations shall be mandatory.

NFPA guidelines to be considered

14.1.1 New fire pumps shall be provided.

Remarks: 12 off main pumps at 795 m3/hr plus 2 jockey pumps at 341 m3/hr

14.1.2 Fire trucks shall be provided.

Remarks: Yes as listed in PDRP4240-8140-SP-1001

14.1.3 A foam system shall be provided.

Remarks: As per requirement

14.1.4 Sprinkler systems shall be provided per OISD guidelines

Locations	Type	Extent
Buildings	Sprinkler	Total
Equipment	Deluge	Required
LPG Vessels	Deluge	Total
Pipe Racks	Not Applicable	
Loading Racks	Not Applicable	
Cone Roof Tanks	Deluge	Roof and Shell
Floating Roof Tanks	Deluge	Shell Only
Other		

Remarks: None

14.1.5 Detection, Alarm and Control Systems shall be provided as follows (include toxic gases as well as fire hazards): As per OISD requirements

Locations	Type	Extent
Buildings	Smoke	Total
Equipment	Heat	Pneumatic T
LPG Vessels	Not Applicable	
Pipe Racks	Not Applicable	
Loading Racks	Not Applicable	
Cone Roof Tanks	Not Applicable	
Floating Roof Tanks	Linear Heat	In Seal
Electrical Sub-stations	Smoke	Total

Remarks: None

Fire and Safety Display System:
[ ] Centralised in Fire Station [ ] Individual in each control room

14.2 Blowdown and Flare System

14.2.1 Blowdown and flare facilities shall be provided.

Remarks: The Facility shall be designed in accordance with design basis PDRP4240-8110-PD-220-0001.

14.2.2 Flares shall be (elevated, smokeless).

Remarks: In accordance with design basis PDRP4240-8110-PD-220-0001.

14.2.3 NO HYDROCARBON SHALL BE VENTED DIRECTLY TO ATMOSPHERE

14.2.4 Relief valves handling hydrocarbon vapour shall be vented to a closed relief system discharging to Flare Stacks.

14.2.5 The liquid relief system shall be combined with the vapour relief system.

Remarks: Liquid relief to the flare system shall be minimised. ISBL KO drums shall be provided to collect liquid reliefs to prevent liquid slugs in main flare line.

14.2.6 The allowable ground level radiation shall be 6.31 W/m^2 for Process Equipment and 3.15 W/m^2 for Refinery boundary [solar excluded].

Remarks: None

14.2.7 The air pollution control requirements are as follows:

A minimum of 7 ambient air quality monitoring stations (SO2, NOx and HC) should be set up in the refinery area in consultation with SPCB, based on occurrence of maximum Ground Level Concentration (GLC) and down wind direction of wind as well as in the direction of Betelvine plantations, the monitoring network must be decided based on modelling exercise to represent short term GLC's. In addition a mobile vehicle with adequate facilities to monitor ambient air quality outside the refinery premises should be provided. Online continuous monitoring system with recording facilities shall be installed in the stacks for various critical parameters such as NOx, CO, SO2 and all Hydrocarbons.

14.3 Aircraft Warning Lights

Warning lights for aircraft shall be provided for structures in excess of 40m

Refer Lighting and Small Power Project Specification, PDRP-8531-SP-0011

14.4 Safety Showers

Safety showers and eye wash stations shall be provided:

Remarks: Quantities and locations to be defined

15 Shipping Facilities

15.1 Marine Facilities

15.1.1 Marine facilities shall be provided.

15.1.2 The facilities required are jetty, topsides including loading arms, piping etc.

Remarks: Jetty By IOCL

15.2 Rail Facilities

15.2.1 No Rail facilities need to be developed inside the refinery Complex.

15.3 Truck Facilities

15.3.1 Truck facilities shall be provided.

15.3.2 The facilities required are (eg. loading rack, weight bridge, scale house, other).

Remarks: Coke, Sulphur and Poly Propylene export

15.4 Ship Security

15.4.1 Shipping Security facilities shall be provided.

15.4.2 The facilities required are (eg. guard house, gates, other).

Remarks: By IOCL

15.5 Pipe Runs

15.5.1 Pipelines shall be provided.

15.5.2 The facilities required are SPM, Stormwater outfall, marine lines, land lines, pump station, metering, etc.

Remarks: SPM by IOCL

16 Environmental

The environmental requirements on this Project are summarized in the Environmental Design Basis (PDRP4280-8150-PH-0002). These requirements include limits on air emissions, effluent discharge, waste and noise. A brief outline of the environmental permitting system in also given in the document.

17 Special Project Requirements

Installation layout shall be per OISD118.